

Lecture 4: Theory of the Firm

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1. The Firm and the Production Function

A firm transforms **inputs** into **outputs**. We focus on the simplest case: n inputs $x \in \mathbb{R}_+^n$ and a single output $y \in \mathbb{R}_+$. The firm's technology is summarized by a production function

$$f : \mathbb{R}_+^n \longrightarrow \mathbb{R}_+, \quad y = f(x).$$

Interpretation: $f(x)$ is the *maximum* output achievable with input vector x . Throughout, the firm's objective is to **maximize profit**.



1.1. Standing Properties of $f(\cdot)$

The production function f is

1. **Continuous** on $\mathbb{R}_+^n$ — small changes in inputs produce small changes in output.
2. **Strictly increasing** — more of any input strictly raises output.
3. **Strictly quasiconcave** on $\mathbb{R}_+^n$ — inputs are at least mildly complementary.
4. Satisfies $f(0) = 0$ — no inputs, no output.

Strict quasiconcavity (a slight strengthening of standard quasiconcavity) gives well-defined first-order conditions and uniquely identified optimal input choices.

2. Key Concepts

2.1. Marginal Product

$$MP_i(x) \equiv \frac{\partial f(x)}{\partial x_i}.$$

The rate at which output increases when one more unit of input i is used, holding all other inputs fixed.

2.2. Isoquants

For an output level $y \geq 0$,

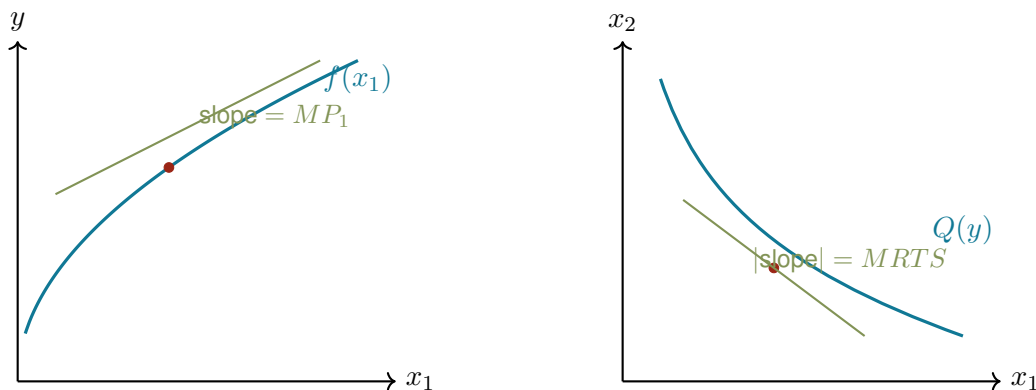
$$Q(y) \equiv \{x \geq 0 \mid f(x) = y\}.$$

The set of input vectors that produce exactly y units of output.

2.3. Marginal Rate of Technical Substitution

$$MRTS_{ij}(x) \equiv \frac{\partial f(x)/\partial x_i}{\partial f(x)/\partial x_j}.$$

The rate at which input i can be substituted for input j while keeping output constant; geometrically, the absolute slope of the isoquant in the (x_i, x_j) plane.



2.4. Returns to Scale

What happens when *all* inputs are scaled by the same factor $t > 0$?

The technology exhibits:

- **Constant returns** if $f(tx) = t f(x)$ for all $t \geq 0$.
- **Increasing returns** if $f(tx) > t f(x)$ for $t > 1$.
- **Decreasing returns** if $f(tx) < t f(x)$ for $t > 1$.

Example

Cobb–Douglas: $f(x_1, x_2) = x_1^{\alpha_1} x_2^{\alpha_2}$ with $\alpha_1, \alpha_2 > 0$.

Marginal product: $\frac{\partial f}{\partial x_i} = \alpha_i x_i^{\alpha_i-1} x_j^{\alpha_j}$.

MRTS: $MRTS_{ij}(x) = \frac{\alpha_i}{\alpha_j} \frac{x_j}{x_i}$.

Returns to scale: $f(tx) = t^{\alpha_1+\alpha_2} f(x)$, so the technology exhibits constant / increasing / decreasing returns according as $\alpha_1 + \alpha_2$ equals, exceeds, or falls below 1.

A single technology can exhibit *different* returns to scale over different ranges. For instance,

$$f(x) = \frac{K}{1 + (x_1^{\alpha_1} x_2^{\alpha_2})^{-1}}$$

displays increasing returns at low input levels and decreasing returns once output approaches the saturation K .

3. Cost Minimization

A profit-maximizing firm must, for any output target y , choose inputs in the cheapest way. We separate this *conditional* cost-minimization problem from the outer profit-maximization problem.

Assumption (price-taking on inputs). The firm takes input prices $w = (w_1, \dots, w_n) \geq 0$ as given.

$$c(w, y) \equiv \min_{x \in \mathbb{R}_+^n} w \cdot x \quad \text{s.t.} \quad f(x) \geq y.$$

The minimum expenditure on inputs required to produce at least y when input prices are w .

If f is continuous, strictly increasing, and strictly quasiconcave, and $w \gg 0$, the cost-minimization problem has a unique solution

$$x^c = x^c(w, y),$$

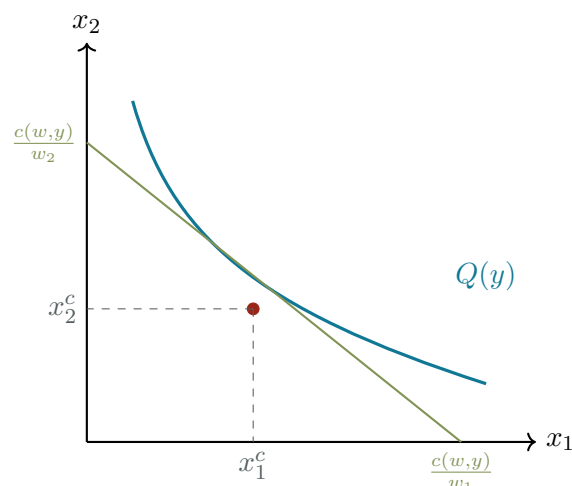
the **conditional input demand**.

3.1. First-Order Conditions

Strict monotonicity makes the constraint $f(x) \geq y$ bind, so we minimize $w \cdot x - \lambda(f(x) - y)$:

$$\frac{\partial \mathcal{L}}{\partial x_i} = w_i - \lambda \frac{\partial f(x)}{\partial x_i} = 0 \quad \implies \quad \frac{w_i}{w_j} = \frac{\partial f(x)/\partial x_i}{\partial f(x)/\partial x_j} = MRTS_{ij}(x)$$

At the optimum, the isoquant is tangent to the isocost line: ratios of input prices equal ratios of marginal products.



3.2. Properties of the Cost Function

The cost function is

1. **Homogeneous of degree 1** in w : $c(tw, y) = t c(w, y)$, $t > 0$.
2. **Strictly increasing** in y .
3. **Concave** in w .
4. **Shephard's Lemma**: if differentiable in w ,

$$\frac{\partial c(w, y)}{\partial w_i} = x_i^c(w, y).$$

3.3. Properties of Conditional Input Demands

1. Own-price effects are non-positive: $\frac{\partial x_i^c(w, y)}{\partial w_i} \leq 0$.
2. Cross-price effects are symmetric: $\frac{\partial x_i^c(w, y)}{\partial w_j} = \frac{\partial x_j^c(w, y)}{\partial w_i}$.
3. $x^c(w, y)$ is homogeneous of degree 0 in w .

3.4. Cobb–Douglas Cost Function

Example

For $f(x) = x_1^{\alpha_1} x_2^{\alpha_2}$, the Lagrangian gives

$$\frac{w_1}{w_2} = \frac{\alpha_1}{\alpha_2} \frac{x_2}{x_1} \implies x_2 = \frac{\alpha_2}{\alpha_1} \frac{w_1}{w_2} x_1.$$

Substituting into $f(x) = y$ yields the conditional demands

$$x_1^c(w, y) = y^{1/(\alpha_1 + \alpha_2)} \left(\frac{w_2}{w_1} \cdot \frac{\alpha_1}{\alpha_2} \right)^{\alpha_2/(\alpha_1 + \alpha_2)},$$

$$x_2^c(w, y) = y^{1/(\alpha_1 + \alpha_2)} \left(\frac{w_1}{w_2} \cdot \frac{\alpha_2}{\alpha_1} \right)^{\alpha_1/(\alpha_1 + \alpha_2)},$$

and the cost function $c(w, y) = w_1 x_1^c + w_2 x_2^c = y^{1/(\alpha_1 + \alpha_2)} \phi(w_1, w_2)$ for an explicit constant ϕ . Cobb–Douglas demands take the convenient form

$$x_i^c(w, y) = \frac{\alpha_i}{w_i} c(w, y).$$

4. Profit Maximization

Assumption (price-taking on output). The firm sells at exogenous output price $p \geq 0$.

$$\pi(p, w) \equiv \max_{x \in \mathbb{R}_+^n} p f(x) - w \cdot x.$$

The solution $x(p, w)$ is the (unconditional) **input demand**, and $y(p, w) = f(x(p, w))$ is the **output supply**.

4.1. Properties of the Profit Function

The profit function is

1. Increasing in p and decreasing in w .
2. **Homogeneous of degree 1** in (p, w) : $\pi(tp, tw) = t \pi(p, w)$, $t > 0$.
3. **Convex** in (p, w) .
4. **Hotelling's Lemma**: if differentiable,

$$\frac{\partial \pi(p, w)}{\partial p} = y(p, w), \quad - \frac{\partial \pi(p, w)}{\partial w_i} = x_i(p, w).$$

5. Price effects on supply and input demand:

$$\frac{\partial y(p, w)}{\partial p} \geq 0, \quad \frac{\partial x_i(p, w)}{\partial w_i} \leq 0.$$

6. Symmetry of cross-effects:

$$\frac{\partial x_i(p, w)}{\partial w_j} = \frac{\partial x_j(p, w)}{\partial w_i}.$$

4.2. Two-Stage View: From Cost to Profit

Cost minimization is the inner stage; profit maximization is the outer stage:

$$\pi(p, w) = \max_{y \geq 0} [p y - c(w, y)].$$

The first-order condition is $p = \partial c(w, y) / \partial y$ — price equals marginal cost. At the optimum,

$$x(p, w) = x^c(w, y(p, w)).$$

4.3. Cobb–Douglas Example

For $f(x) = x_1^{\alpha_1} x_2^{\alpha_2}$, $c(w, y) = y^{1/(\alpha_1 + \alpha_2)} \phi(w_1, w_2)$. The profit problem becomes

$$\max_{y \geq 0} p y - y^{1/(\alpha_1 + \alpha_2)} \phi(w_1, w_2).$$

First-order condition:

$$p = \frac{1}{\alpha_1 + \alpha_2} y^{1/(\alpha_1 + \alpha_2) - 1} \phi(w_1, w_2).$$

Second-order condition:

$$\frac{\partial^2 \pi}{\partial y^2} = - \frac{(\alpha_1 + \alpha_2 - 1)}{(\alpha_1 + \alpha_2)^2} y^{1/(\alpha_1 + \alpha_2) - 2} \phi(w_1, w_2).$$

Three regimes.

- If $\alpha_1 + \alpha_2 < 1$ (decreasing returns), π is strictly concave in y ; the FOC pins down a unique optimum

$$y(p, w) = \left(\frac{p(\alpha_1 + \alpha_2)}{\phi(w_1, w_2)} \right)^{(\alpha_1 + \alpha_2)/(1 - \alpha_1 - \alpha_2)}.$$

- If $\alpha_1 + \alpha_2 > 1$ (increasing returns), π is strictly convex in y ; profit can be made arbitrarily large by scaling up. **No bounded solution.**

- If $\alpha_1 + \alpha_2 = 1$ (constant returns), π is linear: $\pi = y(p - \phi(w_1, w_2))$. Three subcases:

- $p < \phi$: optimum at $y = 0$.
- $p > \phi$: profit unbounded.
- $p = \phi$: any $y \geq 0$ is optimal — indeterminate supply.

Constant returns plus competitive output pricing implies zero economic profit at the optimum. This is the building block for partial-equilibrium analysis in the next lecture.

Reference: Jehle & Reny, Advanced Microeconomic Theory, Chapter 3.